

Farm Intelligence Platform

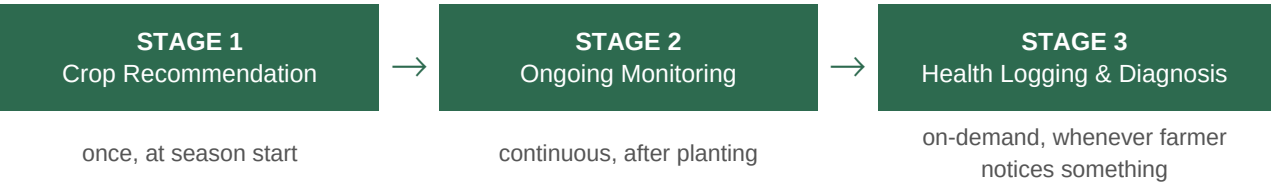
Voice + SMS agricultural advisory in Indic languages, for small and marginal farmers

The Problem

Farmers often lose crops and money because crop choices are based on habit or hearsay, not on real soil, water, or weather data. Once a crop is planted, there is usually no early-warning system to catch dry spells, disease, or pest risk before damage is done.

The Solution — One Engine, Three Stages

All three stages read and write to **one shared record per farmer, per field** — not three separate apps stitched together. Stage 1 sets the baseline, Stage 2 keeps it updated in real time, and Stage 3 lets the farmer correct or confirm it at any point.



STAGE 1 — CROP RECOMMENDATION

Farmer gives (via IVR call or SMS — no smartphone needed):

- District and village name
- Current season (Kharif / Rabi / Zaid)
- Water source type (rainfed / borewell / canal)
- Last crop grown (for crop rotation logic)
- *Optional, if using the app:* exact farm coordinates

System auto-fetches (farmer does nothing here):

- **Soil Health Card** — pH, NPK, organic carbon, texture (district level)
- **CGWB** — groundwater depth (district level)
- **IMD** — seasonal rainfall forecast (district level)
- **Agmarknet** — mandi price trends for candidate crops
- **Satellite check (Sentinel-1 + Sentinel-2)** — sharpens the district-level numbers down to the farmer's exact field, using real soil moisture (radar) and past NDVI crop-health history at that plot

Scoring — safety before profit, always:

Step A — Hard filters. Every candidate crop is checked against soil fit, water fit, and rainfall fit. Any crop that fails is eliminated immediately, even if it is highly profitable — a thirsty cash crop is never recommended to a rainfed farmer with low groundwater.

	Track 1 — Safe & Familiar	Track 2 — Profit-Optimised
Soil fit	30%	30%
Water fit	30%	30%
Rainfall fit	25%	20%
Local familiarity / input availability	15%	—

Time-adjusted ROI / price trend / buyer access	—	20%
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ROI weight is deliberately kept low so a high-margin crop can never override a poor agronomic fit. A crop earning █80,000/acre over 3 years scores lower than one earning █39,000/acre in 70 days, because the farmer needs income this season, not in 3 years.

Output: voice message or SMS, in the farmer's language and script — “Safe options: Crop A, Crop B. For higher profit, Crop C is also worth considering.”

STAGE 2 — ONGOING MONITORING

Runs continuously after planting, combining four data sources that update at different speeds:

- **Sentinel-1 (SAR / radar)** — real soil moisture, works through monsoon clouds, updates every ~6 days
- **Sentinel-2 (optical)** — NDVI / NDWI / NDRE, tracks plant health and water stress, can catch problems before they're visible to the eye, ~5 day revisit
- **Weather forecast (API)** — rainfall, temperature, evapotranspiration: what's *coming*, updated daily
- **IoT ground sensor (optional, tiered)** — real-time soil moisture where available, sharpens the satellite estimate for that exact field

Core logic — Water Balance Projection:

Soil moisture (now)	+	Rainfall (forecast)	–	Evapotranspiration (weather-based)
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↓ Will moisture cross the crop's critical threshold before rain arrives?

YES	Dry-spell alert: “Irrigate within X days”
NO	Hold action, or “rain coming — don't fertilize now”

Output: irrigation timing, fertilization timing, dry-spell early warnings — delivered by voice/SMS.

STAGE 3 — CROP HEALTH LOGGING & DIAGNOSIS

Farmer sends a photo or voice note whenever they notice something wrong in the field.

- If Stage 2's satellite layer already flagged an anomaly in that spot (stress visible before symptoms), the farmer's photo **confirms and identifies** it — satellite can say “something's wrong here,” only the photo can say which disease or pest
- AI diagnosis runs on the photo/voice note
- If unresolved or serious, the case connects directly to a **Rythu Seva Kendra** for expert human follow-up

Feedback loop: a confirmed diagnosis here recalibrates what future satellite anomalies at that same spot are likely to mean — Stage 3 makes Stage 2 smarter over time.

Honest Limits — Said Upfront, On Purpose

Being explicit about what satellite data can and cannot do builds credibility with anyone reviewing this technically, rather than overselling “AI predicts everything.”

Detect water stress	YES — Sentinel-1 (soil) + Sentinel-2 (plant)
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Detect early plant stress, before visible symptoms	YES — NDRE, PRI indices
Detect pests directly	NO — only risk-inference from weather + crop stage + regional outbreak data
Identify the exact disease	NO — only flags “something's wrong here”; farmer's photo does the ID
Check soil microbial health	NO — needs real soil test data (Soil Health Card)

Why This Is One Idea, Not Three Stitched Together

- Stage 1's satellite check and Stage 2's satellite monitoring use the **same two satellites** (Sentinel-1, Sentinel-2) — Stage 1 uses historical data for that plot, Stage 2 uses live, ongoing data
- Stage 3's farmer photo **feeds back** into Stage 2 — a confirmed diagnosis recalibrates what future satellite anomalies at that spot likely mean
- Everything is delivered the same way — **voice / SMS, local language, no smartphone required** — matching the challenge's core requirement for small and marginal farmers